

If you are 18 years old, you have lived at least 157,680 hours. If you live to be 80, you will live another 543,120 hours. This test is 1 hour of your life. Keep your perspective and don't sweat it. Good luck!

Circle True or False. Read each statement completely before answering.

1. **TRUE** False The oldest trace fossils ever discovered were horizontal trace fossils.
2. **TRUE** False The best classification system is that which most closely reflects evolutionary history.
3. **TRUE** False Reinforcement decreases the chance of forming hybrids.
4. True **FALSE** The basal group is the most primitive group on a phylogenetic tree.
5. True **FALSE** An asteroid caused the extinction of all dinosaurs about 65 million years ago.
6. True **FALSE** Reinforcement enhances the chance of forming hybrids.
7. True **FALSE** If populations of a single species are geographically isolated for *any* period of time, then they will evolve biological reproductive isolation.
8. **TRUE** False Identifying cryptic species is a problem with the morphological species.
9. **TRUE** False Endosymbiosis states that mitochondria evolved before chloroplasts.
10. **TRUE** False The species is the *largest* unit within which gene flow can readily occur.

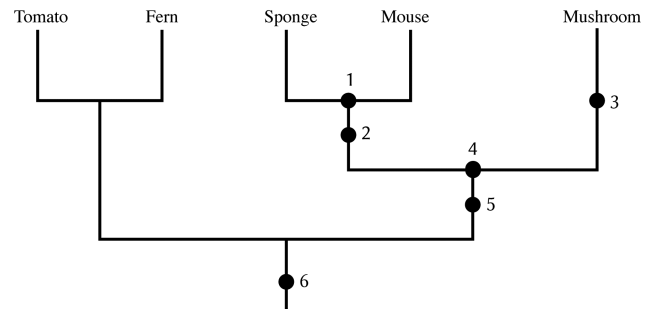
Multiple choice. Choose the *best* answer. Clearly circle the letter for your answer. Vague or unclear marks will be counted as incorrect.

11. Males of different species of fruit fly *Drosophila* that live in the same parts of the Hawaiian Islands have different elaborate courtship rituals. These rituals involve fighting other males and making stylized movements that attract females. What type of reproductive isolation does this represent?
 - A. Mechanical isolation.
 - B. Habitat isolation.
 - C. Behavioral isolation.**
 - D. Gametic isolation.
 - E. Temporal isolation.
 12. A hybrid zone is properly defined as
 - A. a zone where the two closely related species remain distinct through hybrid fusion.
 - B. an area where mating occurs between members of two closely related species, producing at least some viable offspring.**
 - C. an area where members of two closely related species intermingle, but gene flow is prevented by allopatric barriers.
 - D. an area where the ranges of two closely related species overlap, but do not interbreed.
 - E. an area where members of two closely related species intermingle, but gene flow is prevented by prezygotic barriers.
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13. Most insects have two pairs of wings. Flies only have one pair of wings. The second pair of wings is reduced to tiny structures called halteres. The mutation(s) that led to the change from wings to halteres could have
- A. resulted in paedomorphosis.
 - B. caused expression of a pseudogene.
 - C. caused the loss of *Antp* or .
 - D. altered the expression of a *Hox* gene.**
 - E. duplicated all of the *Hox* genes in the fly genome.

14. Which of the six numbered corresponds to the most recent common ancestor of a mushroom and a sponge?

- A. 5.**
- B. 6.
- C. 2.
- D. 1.
- E. 4.
- F. 3.



15. Which of the following factors would *not* contribute to allopatric speciation?
- A. Different mutations begin to distinguish the gene pools of the separated populations.
 - B. One population is exposed to different selection pressures than the other population.
 - C. The separate population is small, and genetic drift occurs.
 - D. Gene flow between the two populations is extensive.**
16. What does the biological species concept use as the primary criterion for determining whether two populations are the same species?
- A. Gene flow.**
 - B. Allopatric speciation.
 - C. Geographic isolation.
 - D. Morphological similarity.
 - E. Ecological differences.
17. Male frogs give calls that attract female frogs to approach and mate. Researchers examined mating calls of closely related tree frogs in South America. If reinforcement is occurring, what would you expect if you compare the calls of the two species in zones of sympatry versus zones of allopatry?
- A. Calls would be about the same in both areas.
 - B. Calls would be more different in areas of sympatry.**
 - C. Calls would show they were cryptic species.
 - D. Calls would be more similar in areas of sympatry.
 - E. Calls would become more similar due to fusion.

18. The most important feature that permits a gene to act as a molecular clock is
- A. a relatively consistent average rate of mutation.**
 - B. that it is a conserved gene.
 - C. a large number of base pairs.
 - D. favored by natural selection.
 - E. a recent gene duplication event.
19. The oldest fossil remains of *Homo sapiens* found so far date from about
- A. 6 million years ago.
 - B. 16,000 years ago.
 - C. 60,000 years ago.
 - D. 200,000 years ago.**
 - E. 1.6 million years ago.
20. Bird guides once listed the Myrtle Warbler and Audubon's Warbler as distinct species. Recently, these birds have been classified as eastern and western forms of the Yellow-Rumped Warbler. Which one of the following pieces of evidence, if true, would be cause for this reclassification?
- A. The two forms interbreed often in nature, and their offspring survive and reproduce well.**
 - B. The two forms live in similar habitats and have similar food requirements.
 - C. The two forms have many genes in common.
 - D. The two forms overlap for a small part of their geographic ranges.
 - E. The two forms are morphologically similar.
21. (5 points) Put the following evolutionary events in order by numbering them from earliest (1) to latest (5).
- a. **4** *Archaeopteryx*
 - b. **5** *Australopithecus*
 - c. **1** Cyanobacteria
 - d. **3** Endosymbiosis
 - e. **2** Oxygen revolution
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Match the *most appropriate* term to each concept. Not all terms are used. No term is used twice.

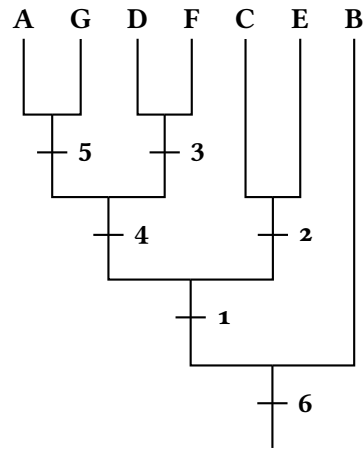
adaptive radiation	dentary-squamosal	<i>Ichthyostega</i>
<i>Archaeopteryx</i>	ecological isolation	mechanical isolation
articular-quadrato	<i>Eusthenoperon</i>	morphological species
articular-squamosal	fusion	natural selection
behavioral isolation	gametic isolation	postzygotic barrier
convergent evolution	habitat isolation	prezygotic barrier
cryptic species	homology	reinforcement
dentary-quadrato	homeotic gene	temporal isolation

22. **Convergent evolution** Unrelated organisms evolve similar traits by adapting to similar habitats.
23. **Fusion** Most likely outcome of a hybrid zone if the hybrid offspring can survive and reproduce as well as the parent species.
24. **Cryptic Species** May be overlooked by the morphological species concept.
25. **Gametic Isolation** Sperm and eggs are incompatible, preventing copulation.
26. **Dentary-Squamosal** The type of jaw joint found in mammals.
27. **adaptive radiation** Evolution of many species in a short period of evolutionary time.
28. *Archaeopteryx* Transitional form between non-flying dinosaurs and modern birds.
29. **Postzygotic Barrier** Any biological process that prevents a hybrid zygote from developing.
30. (4 points) Two species of fishes reproduce at the same time but do not interbreed in the wild. One species lives in the lake, while the other species lives in the streams that flow into the lake. Based only on this information, which specific type of prezygotic or specific type of postzygotic barrier most likely keeps these two fish species reproductively isolated? Explain how this reproductive barrier prevents them from interbreeding.

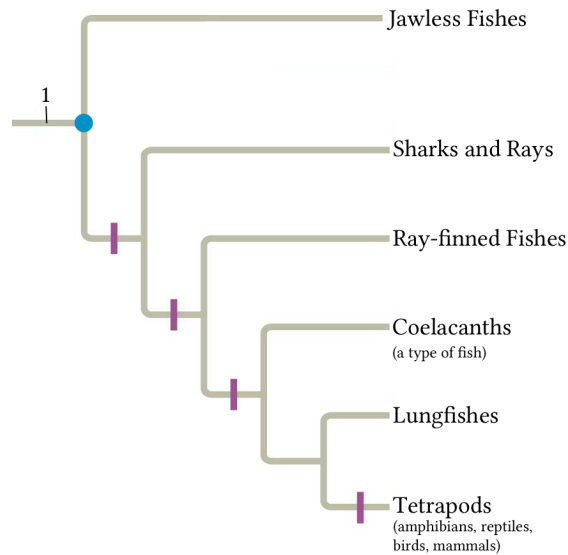
Habitat isolation (2 pts). They occupy different habitats and so never come into contact with each other during reproductive time. (2 pts)
Also acceptable: Gametic isolation (2 pts). The gametes mix in the lake but are incompatible and prevent fertilization. (2 pts)

31. (15 points) The table shows the presence and absence of six homologies (1–6) among seven species (A–G). Use these data to draw a phylogenetic tree. Draw a tick mark across the branch of each common ancestor and label the homology that unites that clade.

	1	2	3	4	5	6
A	1	0	0	1	1	1
B	0	0	0	0	0	1
C	1	1	0	0	0	1
D	1	0	1	1	0	1
E	1	1	0	0	0	1
F	1	0	1	1	0	1
G	1	0	0	1	1	1



32. Use the following phylogenetic tree to answer the following questions.



(a) (5 points) Are the coelacanths more closely related to ray-finned fishes or to tetrapods. Tell how you know.

Coelacanths are more closely related to tetrapods (2 pts) because they share a more recent common ancestor with tetrapods than ray-finned fishes (2 pts).

(b) (5 points) Which group is a basal group (taxon)? What does this mean about its evolution compared with that of the other groups?

The jawless fishes are basal (2 pts). That means these fishes evolved before the other groups of fishes and tetrapods (2 pts).

(c) (5 points) “Fishes” includes all groups shown above except for tetrapods. Are fishes a monophyletic group? Tell how you know.

Fishes are not monophyletic (2 pts). Fishes contain some but not all of the descendants of the common ancestor (2 pts).

(d) (5 points) What does the branch labeled “1” represent on this tree? What is the proper term for this branch?

The root (1 pt) represents the common ancestor of all groups shown on the phylogeny (2 pts). Grade only for 3 points.